

CANDIDATE NAME _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE

JC2 Preliminary Examination 2013

H1 BIOLOGY

Paper 2

8875
18 August 2013, Monday

2 Hours

Additional materials:

Writing papers / OMS

Date / Day:

INSTRUCTIONS TO CANDIDATES

Write your name, CG and index number in the spaces at the top of this page and on all separate writing papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

Section B

Answer **only one** question out of two.

Write your answers on the separate answer paper provided.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
1	/6
2	/8
3	/5
4	/10
5	/11
Section B	
6	/20
7	/20
TOTAL	/60

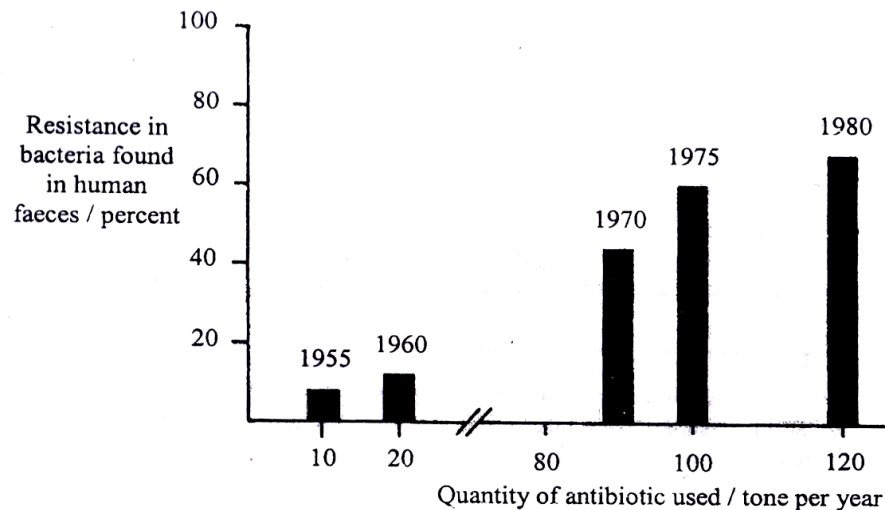
This question paper consists of **11 printed pages and 1 blank page**

Question 1

Antibiotic medications are used to kill bacteria, which can cause illness and disease. They have made a major contribution to human health. Many diseases that once killed people can now be treated effectively with antibiotics. However, some bacteria have become resistant to commonly used antibiotics.

Antibiotic resistant bacteria are bacteria that are not controlled or killed by antibiotics. Due to mutation, they are able to survive and even multiply in the presence of an antibiotic. Most infection-causing bacteria can become resistant to at least some antibiotics. Bacteria that are resistant to many antibiotics are known as multidrug-resistant organisms (MROs).

The bar chart below shows the levels of resistance of human faecal bacteria to antibiotics in relation to the amounts of antibiotics used in each of a number of years from 1955 to 1980.



a) Describe the relationship between the incidence of resistance and the use of antibiotics from 1955 to 1980. [2]

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b) Explain how natural selection has led to the emergence of such antibiotic resistant bacteria. [4]

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[Total: 6 marks]

Question 2

Figure 2.1 shows a haemoglobin molecule.

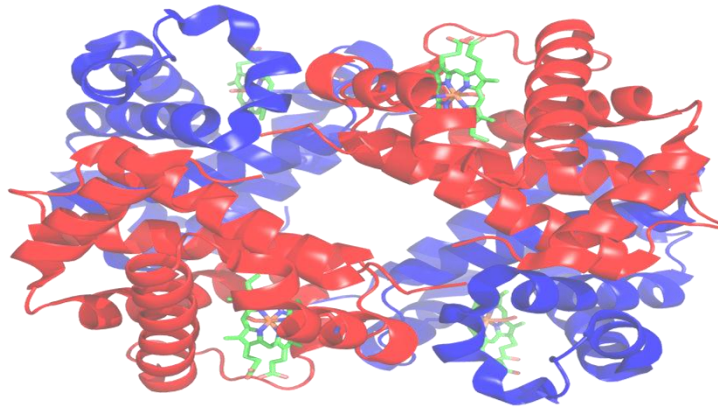


Figure 2.1

a) Discuss how the structure of haemoglobin is related to its function in a mammal. [3]

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b) Haemoglobin is a globular protein. State 2 differences between fibrous and globular proteins. [2]

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c) Predict what will happen to the structure of haemoglobin at high temperature and explain your answer. [3]

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[Total: 8 marks]

Question 3

A man and his wife have blood group AB. However, their first boy has blood group O. Upon consultation with a geneticist, they realised that this is due to the boy lacking the H antigen. This antigen is an essential precursor of the ABO blood group antigens. The boy is homozygous recessive at the H gene locus on chromosome 19. Hence, he cannot synthesise either the A or B antigen on his red blood cells even if he may have the I^A or I^B allele at the ABO gene locus on chromosome 9.

Using suitable symbols, draw a genetic diagram in the space provided below to show the expected blood group phenotypic ratio in the offspring of this couple. [5]

[Total: 5 marks]

Question 4

Rubisco is an important enzyme necessary for photosynthesis. One characteristic of this enzyme is its affinity with oxygen or carbon dioxide, leading to photorespiration and photosynthesis respectively. An investigation to determine the effect of oxygen to carbon dioxide ratio on the amount of carbohydrates produced was carried out. **Table 4.1** tabulates the data collected.

Experiment No.	Light intensity / lux	Amount of CO ₂ available in stroma / %	Amount of O ₂ available in stroma / %	pH	Amount of carbohydrates produced / arbitrary units
1	20	10	0.002	3	0.01
2	20	10	0.002	8	250
3	20	2	0.009	3	0.01
4	20	2	0.009	8	15

Table 4.1

- a) With reference to **Table 4.1**, describe the effect of oxygen on the amount of carbohydrates at pH8. [1]

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- b) Suggest a reason for the low amount of carbohydrates produced at pH3 even though the oxygen to carbon dioxide ratio is altered in experiment numbers 1 and 3. [2]

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c) Explain the effect of high oxygen to carbon dioxide ratio on the amount of carbohydrates produced. [3]

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d) Explain the importance of oxygen in photorespiration. [4]

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[Total: 10 marks]

Question 5

When threatened, lizards lose their tails as a distraction to predators. The detached tail is able to move involuntarily due to muscle spasms for up to five minutes. The process of a lizard breaking off its tail is known as autotomy. The good news for the lizard is that it is able to regenerate its tail in less than a year as shown in **Figure 5.1**.

Interested to find out more about the regenerative ability of the lizard tail, scientists have isolated fast dividing cells in a regenerating lizard tail for more analysis.

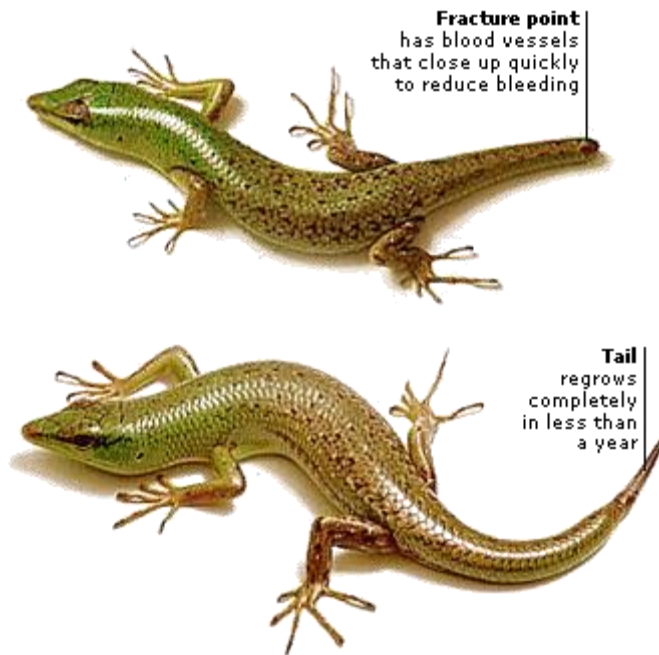


Figure 5.1

- a) State the cellular process that is responsible for the regeneration of the lizard tail. [1]

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- b) List four properties of the cells responsible in the regeneration of the lizard tail. [4]

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c) Contrast the cellular process stated in **a)** to reductive nuclear division. [4]

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Most somatic cells in the human body, unlike the cells from the regenerating lizard tail, have a limited lifespan. However, certain changes to critical genes can occur over time to lead up to the formation of tumors in the body which can then become cancerous.

d) Outline briefly how a tumor becomes cancerous. [2]

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[Total: 11 marks]

Section B (20 marks)

Free-response question

Write your answer on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate

Question 6

- a) Using a named example, describe how large quantities of a particular protein may be produced using bacteria systems. [10]
- b) Describe the advantages and limitations of the Polymerase Chain Reaction (PCR). [4]
- c) Explain how the structures of starch and cellulose are linked to their respective functions. [6]

Question 7

- a) Describe how the structure of the membrane affects the movement of substances into and out of the cell. [8]
- b) Describe the main ways in which an enzyme differs from DNA. [5]
- c) Relate the structure of DNA to its function. [7]

End of Paper

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